ANALYSIS REPORT

Wave 1 Periodic MLP Explicit Harmonic Tracking Campaign Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the approved `Wave 1` periodic MLP explicit harmonic tracking campaign. The campaign tested whether fixed sparse and dense harmonic feature dictionaries improve the existing `periodic_mlp` family without changing the pure `feedforward` baseline or redefining the future `Fourier-Feature MLP` family.

The campaign completed all 9 planned runs with zero launcher failures. The campaign winner by the configured selection policy, minimum `test_mae` with `test_rmse`, `val_mae`, and parameter count as tie breakers, is `te_periodic_mlp_dense240_tracking_Fw`.

Campaign Artifacts

Artifact	Path
Campaign output	`output/training_campaigns/2026-05-20-23-14-17_wave1_periodic_mlp_explicit_harmonic_tracking_campaign_2026_05_20_22_42`
Campaign leaderboard	`output/training_campaigns/2026-05-20-23-14-17_wave1_periodic_mlp_explicit_harmonic_tracking_campaign_2026_05_20_22_42/campaign_leaderboard.yaml`
Best-run pointer	`output/training_campaigns/2026-05-20-23-14-17_wave1_periodic_mlp_explicit_harmonic_tracking_campaign_2026_05_20_22_42/campaign_best_run.yaml`
Execution report	`output/training_campaigns/2026-05-20-23-14-17_wave1_periodic_mlp_explicit_harmonic_tracking_campaign_2026_05_20_22_42/campaign_execution_report.md`
Planning report	`doc/reports/campaign_plans/wave_1/2026-05-20-22-42-49_wave1_periodic_mlp_explicit_harmonic_tracking_campaign_plan_report.md`
Technical document	`doc/technical/2026-05/2026-05-20/2026-05-20-22-34-11_periodic_mlp_explicit_harmonic_basis.md`

Execution Summary

Field	Value
Campaign name	`wave1_periodic_mlp_explicit_harmonic_tracking_campaign_2026_05_20_22_42_49`
Started at	`2026-05-20T23:14:17`
Finished at	`2026-05-21T09:38:37`
Completed runs	9
Failed runs	0
Tested model type	`periodic_mlp`
Tested direction scopes	`global`, `Fw`, `Bw`

Field	Value
Tested harmonic banks	`RCIM sparse` , `0..240` , `0..360`

Campaign Winner

Field	Value
Run name	`te_periodic_mlp_dense240_tracking_Fw`
Run instance	`2026-05-21-08-22-25__te_periodic_mlp_dense240_tracking_fw`
Model family	`periodic_mlp_fw`
Model type	`periodic_mlp`
Direction scope	forward only
Harmonic bank	`0..240`
Trainable parameters	87,681
Validation MAE	0.002541
Validation RMSE	0.003049
Test MAE	0.003055
Test RMSE	0.003537

The winning checkpoint is stored at `output/training_runs/periodic_mlp_fw/2026-05-21-08-22-25__te_periodic_mlp_dense240_tracking_fw/checkpoints/periodic_mlp-epoch=039-val_mae=0.00254077.ckpt` .

Leaderboard

Rank	Run	Scope	Harmonics	Test MAE	Test RMSE	Params
1	`te_periodic_mlp_dense240_tracking_Fw`	Fw	`0..240`	0.003055	0.003537	87,681
2	`te_periodic_mlp_rcim_sparse_tracking_Fw`	Fw	RCIM sparse	0.003131	0.003578	28,545
3	`te_periodic_mlp_dense360_tracking_Fw`	Fw	`0..360`	0.003155	0.003680	118,401
4	`te_periodic_mlp_rcim_sparse_tracking_global`	global	RCIM sparse	0.003275	0.003726	28,545
5	`te_periodic_mlp_dense240_tracking_global`	global	`0..240`	0.003348	0.003862	87,681
6	`te_periodic_mlp_rcim_sparse_tracking_Bw`	Bw	RCIM sparse	0.003398	0.003922	28,545
7	`te_periodic_mlp_dense360_tracking_global`	global	`0..360`	0.003401	0.003831	118,401
8	`te_periodic_mlp_dense240_tracking_Bw`	Bw	`0..240`	0.003417	0.004005	87,681
9	`te_periodic_mlp_dense360_tracking_Bw`	Bw	`0..360`	0.003424	0.004006	118,401

Technical Interpretation

The strongest new result is forward-only `periodic_mlp` with the dense `0..240` fixed periodic-feature bank. It reaches test MAE 0.003055, ahead of the forward RCIM sparse candidate at 0.003131 and the forward dense `0..360` candidate at 0.003155.

The result does not create a universal improvement for every direction scope. The global campaign candidates remain behind the previous global Optuna `periodic_mlp` winner, and the backward campaign candidates remain behind the previous backward Optuna winner. This suggests that simply increasing the fixed harmonic dictionary is useful in the forward-only surface but is not a drop-in replacement for the earlier tuned compact periodic MLP surfaces.

The dense `0..240` forward model is also much larger than the compact Optuna periodic MLP baseline. It should therefore be treated as a stronger curve-fidelity candidate, not as an automatic deployment promotion. The TE Curve Verification Pipeline is still required to decide whether the extra harmonic inputs recover visible TE oscillations or only improve scalar error locally.

Registry Effects

Registry Scope	Current Family Best	Test MAE	Interpretation
`periodic_mlp`	`te_periodic_mlp_h04_standard_global_optuna_t0010`	0.003186	Previous Optuna best remains ahead
`periodic_mlp_fw`	`te_periodic_mlp_dense240_tracking_Fw`	0.003055	Updated by this campaign
`periodic_mlp_bw`	`te_periodic_mlp_h04_standard_Bw_optuna_t0006`	0.003233	Previous Optuna best remains ahead

The current program-level winner remains the previously registered `tree_fw` run with test MAE 0.002743. This closeout therefore does not promote a new program-level best model for deployment.

Closeout Decision

The campaign is complete and successful from an execution standpoint: all 9 runs completed, the leaderboard and best-run artifacts exist, and the relevant family registries were refreshed.

From a modeling standpoint, the explicit harmonic periodic-feature extension is worth keeping. The forward dense `0..240` result is the main candidate to carry into visual validation. The global and backward high-order periodic MLP candidates should not replace the existing compact Optuna-selected family bests unless later curve-level evidence justifies the larger feature bank.

Recommended Follow-Up

1. Generate TE Curve Verification Pipeline curve overlays for the forward dense `0..240`, forward RCIM sparse, and forward dense `0..360` periodic MLP candidates.
2. Compare the same curves against the existing `tree_fw` program winner and earlier compact `periodic_mlp_fw` Optuna baseline.
3. Promote no periodic MLP checkpoint until the overlay review confirms real oscillation tracking rather than scalar-only improvement.
4. Keep the future `Fourier-Feature MLP` family separate from this fixed engineered-feature `periodic_mlp` extension.